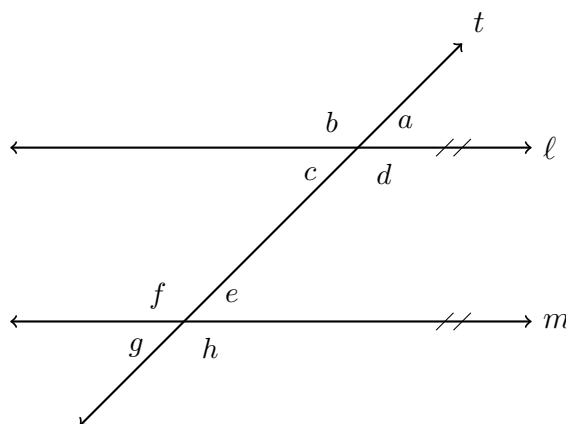


# Angle Measures on Parallel Lines

*Finding an unknown angle from one that is known, solving for  $x$  when a parallel-line angle pair is congruent, and solving for  $x$  when the pair adds to 180 degrees*

The diagram every problem uses.



*Lines  $\ell$  and  $m$  are parallel, and  $t$  is a transversal cutting both. Every problem on this sheet names its angles from this one diagram. No measures are shown here — use the values given in each problem, and do not read anything off the drawing. (Not to scale.)*

**How to work these problems.**

- Decide first whether the two angles named are **congruent** (corresponding, alternate interior, alternate exterior, or vertical) or **supplementary** (a linear pair, or same-side interior). That decision is the whole problem.
- Congruent means *set the expressions equal*. Supplementary means *add them and set the sum to 180*.
- When a problem gives expressions in  $x$ , finding  $x$  is only half the job — substitute it back to get the angle the question asked for.
- Name the relationship you used beside each answer.

1. The measure of  $\angle a$  is  $118^\circ$ . Find the measures of  $\angle d$  and  $\angle e$ .

$$m\angle d = \underline{\hspace{2cm}} \quad m\angle e = \underline{\hspace{2cm}}$$

- 2.** The measure of  $\angle c$  is  $47^\circ$ . Find the measures of  $\angle f$  and  $\angle g$ .

$$m\angle f = \underline{\hspace{2cm}} \quad m\angle g = \underline{\hspace{2cm}}$$

- 3.**  $m\angle b = (3x + 10)^\circ$  and  $m\angle f = (5x - 24)^\circ$ . Find  $x$ , then find  $m\angle b$ .

$$x = \underline{\hspace{2cm}} \quad m\angle b = \underline{\hspace{2cm}}$$

- 4.** The measure of  $\angle h$  is  $125^\circ$ . Find the measures of  $\angle b$  and  $\angle e$ .

$$m\angle b = \underline{\hspace{2cm}} \quad m\angle e = \underline{\hspace{2cm}}$$

5.  $m\angle d = (7x - 15)^\circ$  and  $m\angle f = (4x + 21)^\circ$ . Find  $x$ , then find  $m\angle d$ .

$$x = \underline{\hspace{2cm}} \quad m\angle d = \underline{\hspace{2cm}}$$

6. The measure of  $\angle f$  is  $108^\circ$ . Find the measures of  $\angle d$  and  $\angle a$ . This one takes two relationships, not one.

$$m\angle d = \underline{\hspace{2cm}} \quad m\angle a = \underline{\hspace{2cm}}$$

7.  $m\angle c = (4x + 12)^\circ$  and  $m\angle f = (6x + 8)^\circ$ . Find  $x$ , then find  $m\angle c$ .

$$x = \underline{\hspace{2cm}} \quad m\angle c = \underline{\hspace{2cm}}$$

8.  $m\angle d = (5x - 8)^\circ$  and  $m\angle e = (3x + 12)^\circ$ . Find  $x$ , then find  $m\angle d$ .

$x =$  \_\_\_\_\_  $m\angle d =$  \_\_\_\_\_

9.  $m\angle a = (2x + 34)^\circ$  and  $m\angle g = (5x - 11)^\circ$ . Find  $x$ , then find  $m\angle a$ .

$x =$  \_\_\_\_\_  $m\angle a =$  \_\_\_\_\_

**10.**  $m\angle b = (9x - 4)^\circ$  and  $m\angle e = (3x + 16)^\circ$ . This time nobody tells you which rule applies: decide for yourself whether the two angles are congruent or supplementary, then find  $x$  and  $m\angle b$ . Below your work, write one or two sentences naming the relationship you used and explaining how you knew it was that one rather than the other.

$$x = \underline{\hspace{2cm}} \qquad m\angle b = \underline{\hspace{2cm}}$$